Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **data-intensive services**, APIs, ETL workflows, React interfaces, automated tests, cloud deployments, and production-ready analytics systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **multi-tenant data services** for 500+ users, translating stakeholder workflows into APIs, normalized schemas, monitored jobs, validation logic, documentation, and production releases across cloud storage systems reliably.
- Improved **database-backed platform performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while supporting reliable user-facing analytics workflows daily.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform API and ETL services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** across integrations, release reviews, and operational handoffs for data workflows.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** while reducing release regressions during distributed deployments and checks.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **modular business services** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, integrations, monitoring, debugging, validation, and release workflows for users daily.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, and shared-environment handoffs for engineering teams and reviewers.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **backend APIs and React tooling** handling 10,000+ daily requests, implementing file ingestion, concurrent processing, AI-assisted workflows, automated checks, documented fixes, and sprint delivery for instructors.
- Partnered with users to debug **production workflow defects** by reproducing OCR, persistence, and execution failures, documenting regression cases, reviewing fixes, and cutting feedback turnaround by **60%** safely.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **evaluation pipelines** across 100,000+ records, producing structured metrics, logs, and visual summaries for AI/ML benchmarking, reproducible experiments, model-analysis workflows, researcher-facing reviews, diagnostics, summaries, and analysis.
- Documented **prompt, evaluation, and reporting workflows** for researchers, turning ad hoc experiments into repeatable processes with clearer technical handoffs, validation outputs, team-facing reporting, and follow-up analysis notes.

Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C++, Bash, HTML/CSS
Backend: REST APIs, OOP, service design, validation, Node.js, Express.js exposure, workflow orchestration, documentation
Data: PostgreSQL, MySQL, SQL Server, MongoDB, ETL, indexing, query tuning, analytics logs, data pipelines
Cloud/DevOps: AWS Glue/S3, Docker, Jenkins, CI/CD, Git, GitHub, Kubernetes concepts, Agile, production support
Frontend/AI: React, Next.js, TypeScript UIs, AI-assisted coding, LLM workflows, prompt evaluation, model diagnostics

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services, TypeScript interfaces, Docker workflows, Git collaboration, and reviews.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Engineered a **cloud-based ingestion and indexing pipeline** processing 50K+ records, implementing incremental updates, metadata persistence, semantic retrieval, and performance tuning to reduce retrieval latency by **40%** reliably.

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*

- Designed an **AI-powered data ingestion pipeline** using Transformers and MongoDB, enabling semantic search, premium billing, metadata indexing, and intelligent retrieval across 100K+ indexed image embeddings efficiently online.